

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2016

Mathematics

Paper 1

Higher Level

Time: 2 hours, 30 minutes

300 marks

School stamp

Running total	
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For examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4		14	
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 14 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You will lose marks if all necessary work is not clearly shown.

You may lose marks if the appropriate units of measurement are not included, where relevant.

You may lose marks if your answers are not given in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1

(Suggested maximum time: 10 minutes)

- (a)** During a discussion in Maths class, Adam said:
“Prime numbers are a subset of Natural numbers ($\mathbb{N}$).”

1 2 3 4 5
6 7 8 9 0

Is Adam correct? Explain your answer.

[illegible]

- (b)** Coins in the euro currency are issued in eight different denominations.



- (i) Jennifer has one coin of each denomination in her pocket. How much money does she have in total?

[illegible]

- (ii) Jennifer would like the total value, in cent, of the coins she has in her pocket to be a prime number. What is the least amount of money she would have to add to her total to make it a prime number?

[illegible]

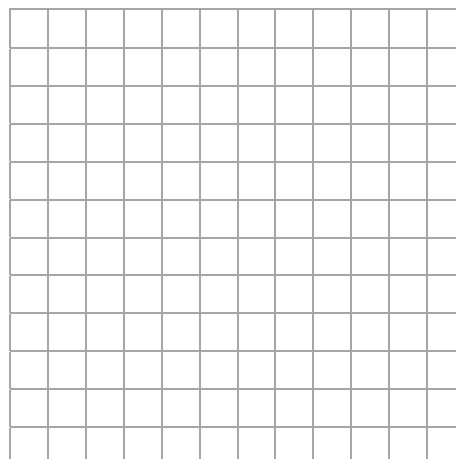
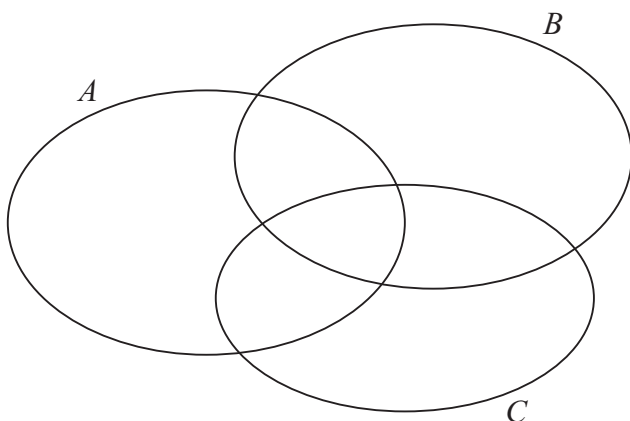
- (c) Express 232 and 368 as a product of prime numbers.
Hence, or otherwise, find the highest common factor of 232 and 368.

[illegible]

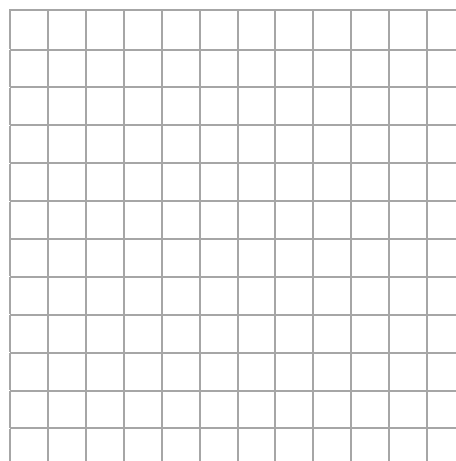
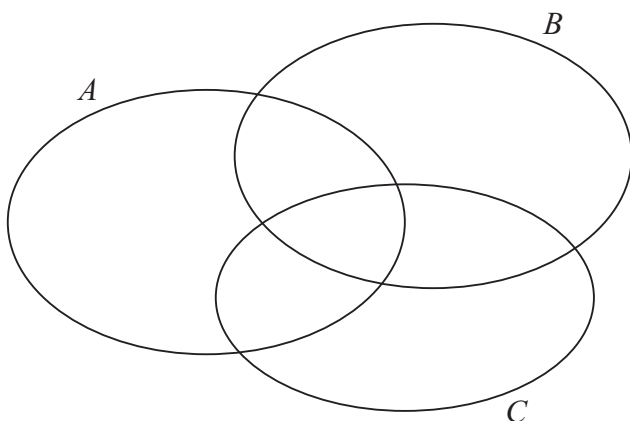
Question 2

(Suggested maximum time: 10 minutes)

- (a) (i) On the Venn diagram below, **shade in** the region that represents $A \setminus (B \cup C)$.
Use set notation to give an alternative description of your shaded region.



- (ii) On the Venn diagram below, **shade in** the region that represents $(A \setminus C) \cap (B \setminus C)$.
Use set notation to give an alternative description of your shaded region.



- (b) A group of 100 students were asked which of the following social networking websites they used in the previous week: Facebook (F), Instagram (I) or Snapchat (S). These are the results:

60 had used Facebook.

45 had used Instagram.

31 had used Snapchat.

12 had used **all three** websites.

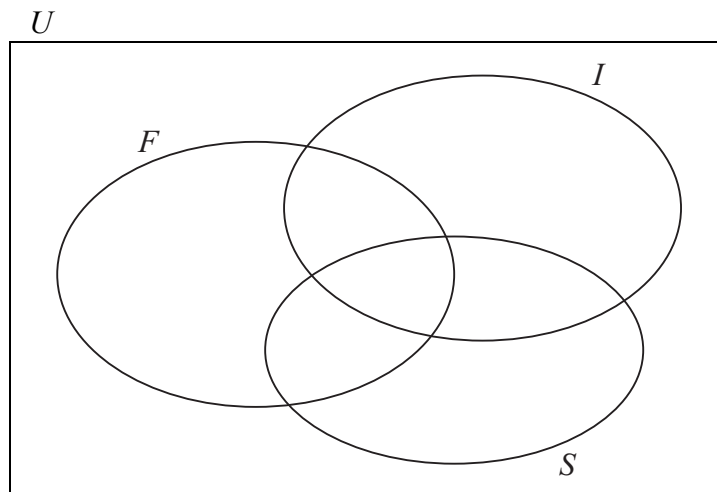
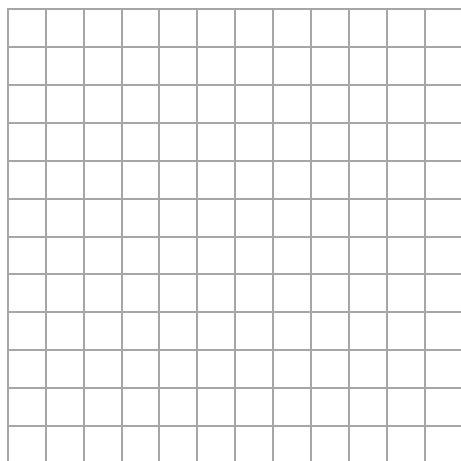
5 had used Facebook **and** Instagram, but **not** Snapchat.

8 had used Facebook **and** Snapchat, but **not** Instagram.

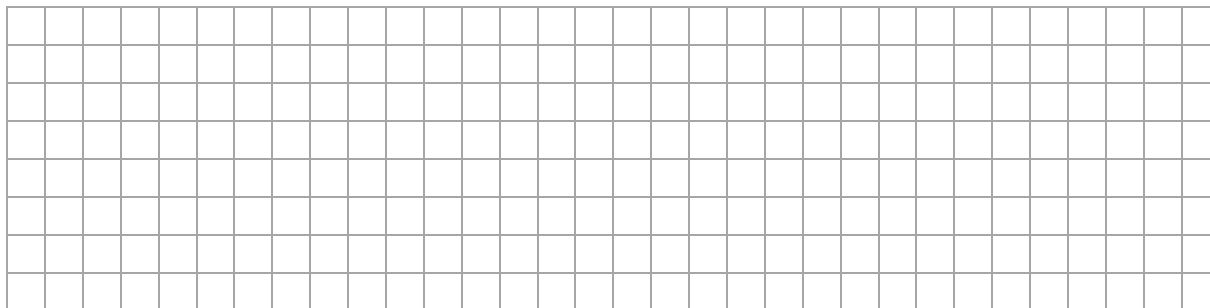
16 had used **exactly two** of the websites listed.



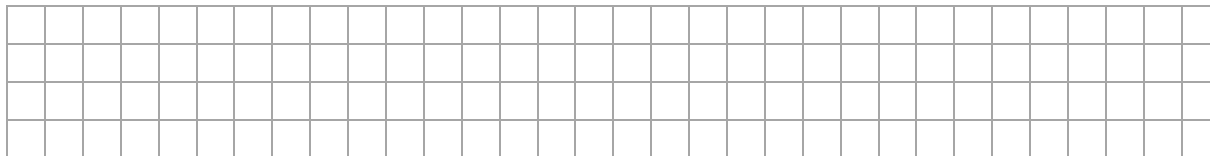
- (i) Represent the above information on the Venn diagram.



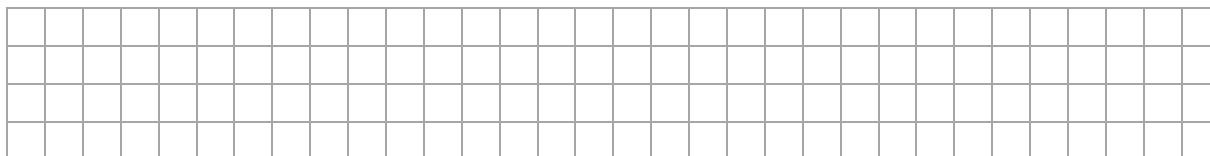
- (ii) How many students did not use any of the social networking websites listed above?



- (iii) Find the **probability** that a student chosen at random from the group used Snapchat only.



- (iv) Find the **probability** that a student chosen at random from those who used the social networking websites listed above used at least two of the websites.



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Question 3

(Suggested maximum time: 15 minutes)

Last summer, Roisin and Tom attended a Young Entrepreneurs camp for transition year students.

- (a) The ratio of girls to boys who attended the camp was 5:3. If there were 60 more girls than boys, find the total number of students who attended the camp.

A large grid of graph paper with 20 columns and 15 rows. The grid is composed of small squares, with a slightly larger margin at the top for writing.

- (b)** As part of the camp, Roisin and Tom set up a small business enterprise selling cupcakes. They made 300 cupcakes and sold them at a local farmers' market.



- (i) Each cupcake cost 35 cent to make and they planned to sell them for €1.40 each. Find the planned **mark-up** (profit as a percentage of cost price) on each cupcake.

[illegible]

- (ii) A stall at the farmers' market cost €40 to rent and the packaging for all the cupcakes cost €15. Find the **total cost** of making and selling all the cupcakes.

A full page of blank graph paper with a uniform grid of small squares. The grid consists of 20 columns and 15 rows, providing a structured area for drawing or writing.

- (iii) On the day of the farmers' market, Roisin and Tom sold 240 cupcakes at the planned price. Towards the end of the day, they sold the remaining cupcakes at a reduced price to ensure that everything was sold. Given that the **overall profit** on the day's enterprise was €226, find the **total** sale price of the remaining cupcakes.

A full-page grid of small squares, typical of graph paper. The grid consists of 20 columns and 15 rows of squares. The lines are thin and gray, creating a uniform pattern across the entire page.

- (iv) The remaining cupcakes were packed together in equal amounts and sold in bundles. In how many different ways could the cupcakes be packed together?



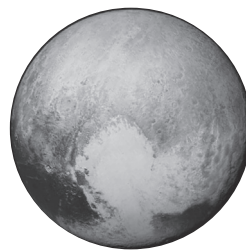
- (v) Roisin and Tom sold the bundles of cupcakes for €5 each. Find the number of cupcakes in each bundle.

A large grid of graph paper with 20 columns and 10 rows. The grid is composed of small squares, with a slightly larger square at the top left corner, likely for a title or header. The grid is empty and ready for use.

Question 4 (Suggested maximum time: 5 minutes)

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On 14 July 2015, the *New Horizons* space probe became the first spacecraft to perform a flyby study of Pluto. During its mission, the space probe took detailed measurements and observations of the planet and its moons.



It is estimated that the volume of Pluto is 6 970 000 000 km³.

- (a) The volume of the Earth is $1.08321 \times 10^{12} \text{ km}^3$. How many times greater than the volume of Pluto is the volume of the Earth? Give your answer correct to the nearest whole number.

[illegible]

- (b)** Find the difference between the diameters of the two planets, in kilometres. Give your answer in the form $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{Z}$, correct to three significant figures.

This image shows a full page of blank graph paper. The grid consists of thin, light gray horizontal and vertical lines that intersect to form small squares across the entire surface. There are no margins, text, or other markings on the paper.

Question 5 (Suggested maximum time: 5 minutes)

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Jim invests a sum of money for one year in a deposit account which offers an annual compound interest rate of 1.5%.

- (a) At the end of the year, the investment is worth €5582.50. Find the sum of money that Jim invested.

A full-page view of a blank sheet of graph paper. The grid consists of small, uniform squares formed by thin, light gray lines. There are no margins, text, or other markings on the page.

At the end of the year, Jim is offered a higher fixed rate of interest if he retains the money in the account for a further two years. Assuming he does not withdraw any money, Jim's investment will be worth €5865.11 at the end of this period.

- (b)** Calculate the annual rate of interest that Jim is offered on his investment. Give your answer as a percentage, correct to one decimal place.

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Question 6

(Suggested maximum time: 10 minutes)

- (a)** Express $\sqrt{32}$ as a power of 2.

[illegible]

- (b)** Hence, solve the equation $\frac{2^5}{2^x} = \sqrt{32}$.

[illegible]

- (c) Verify your answer to part (b).

[illegible]

- (d)** Use the properties of surds to show that $\frac{\sqrt{128} - \sqrt{50}}{\sqrt{8}}$ simplifies to a constant.

[illegible]

Question 7

(Suggested maximum time: 10 minutes)

- (a)** Factorise fully $2x^2 - 6x$.

[illegible]

- (b)** Factorise fully $3a^2c - 9a + 6c - 2ac^2$.

A large grid of graph paper with 20 columns and 10 rows. The grid is composed of small squares, with a slightly larger square at the top left corner, likely for a title or header. The grid is empty and ready for use.

- (c) Use factors to simplify the following: $\frac{2x^2 + 5x - 12}{4x^2 - 9}$.

- (d)** Multiply out and simplify $(x^2 - 3x + 5)(x - 7)$.

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Question 8

(Suggested maximum time: 5 minutes)

- (a) *EuroMillions* is a lottery game operated in several European countries. The winners of the largest lottery jackpot to date are Adrian and Gillian Bayford from Great Britain, who won €190 million on 10th August, 2012.



Given that the exchange rate was €1 = £0.7824 sterling on that day, calculate the value of the jackpot in sterling.

A full-page sheet of white graph paper with a light gray grid. The grid consists of small squares, approximately 10 units wide by 10 units high, covering the entire area of the page.

- (b)** This was not the largest sterling amount won, as Chris and Colin Weir, also from Great Britain, scooped a €185 million jackpot on 12th July, 2011 but received £12·997 million more than the Bayfords due to a more favourable exchange rate.

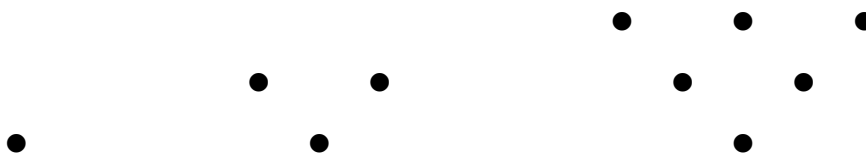
Calculate the exchange rate on that day in the form €1 = £□.□□.

[illegible]

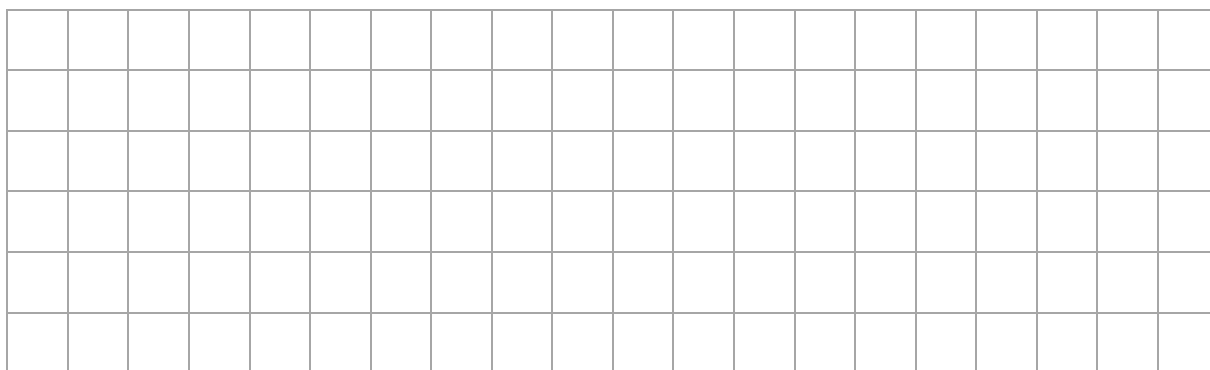
Question 9

(Suggested maximum time: 10 minutes)

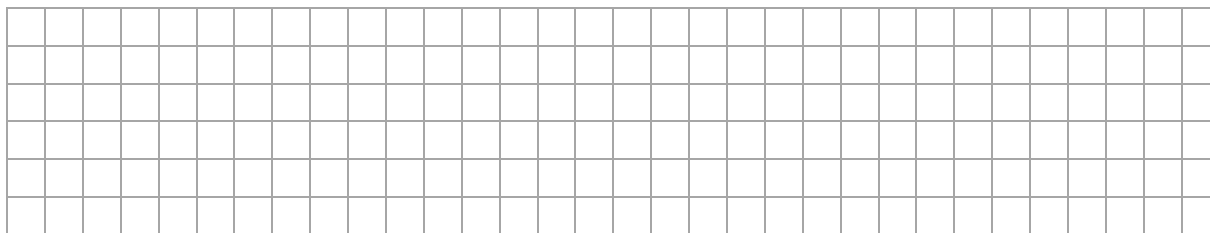
The first three stages of a pattern are shown below.
Each stage of the pattern is made up of bowling pins.



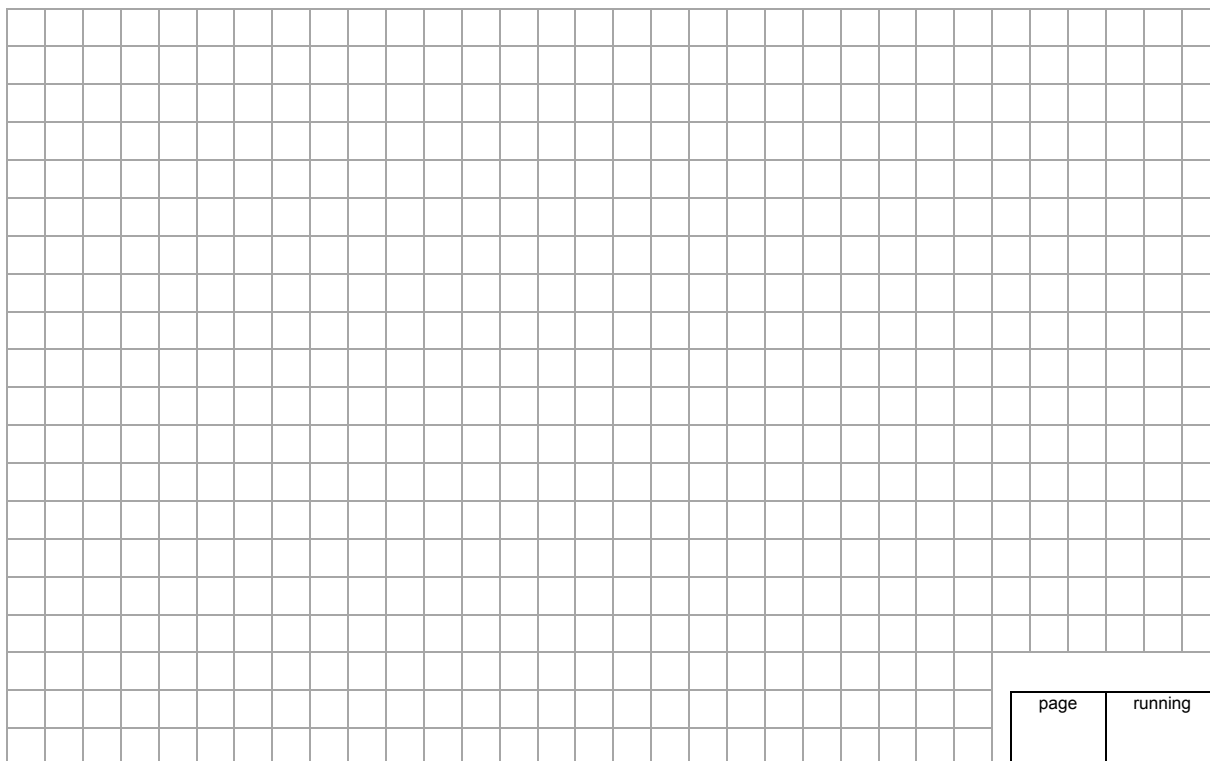
- (a) Draw the next two stages of the pattern.



- (b) Show that the numbers of bowling pins in each stage form a quadratic sequence.



- (c) Find a general formula for the number of bowling pins in stage n of the pattern, where $n \in \mathbb{N}$.



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Question 10 (Suggested maximum time: 15 minutes)

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- (a)** Write the following as a single fraction in its simplest form.

$$\frac{3}{2x-1} - \frac{4}{3x+2}$$

- (b)** x is a real number.
Number A is equal to 1 less than x .
Number B is equal to 2 greater than seven times x .
- (i)** Write down the numbers A and B , in terms of x .

[illegible]

- (ii) The product of these two numbers is 2.
Use this information to write an equation in x .

[illegible]

- (iii) Solve this equation to find the two possible values of x .
Give each of your answers correct to two decimal places.

Question 11

(Suggested maximum time: 10 minutes)

- (a)** Let g be the function $g : x \mapsto 2x^2 - 7$, where $x \in \mathbb{R}$.

- (i) Find the value of $g(-2)$.

[illegible]

- (ii) Find the values of k for which $g(k) = 25$.

[illegible]

- (b) Let $F = \frac{4(4 - C)}{2C + 1}$.

- (i)** Calculate the value of F when $C = \frac{1}{2}$.

[illegible]

- (ii) Write C in terms of F .

[illegible]

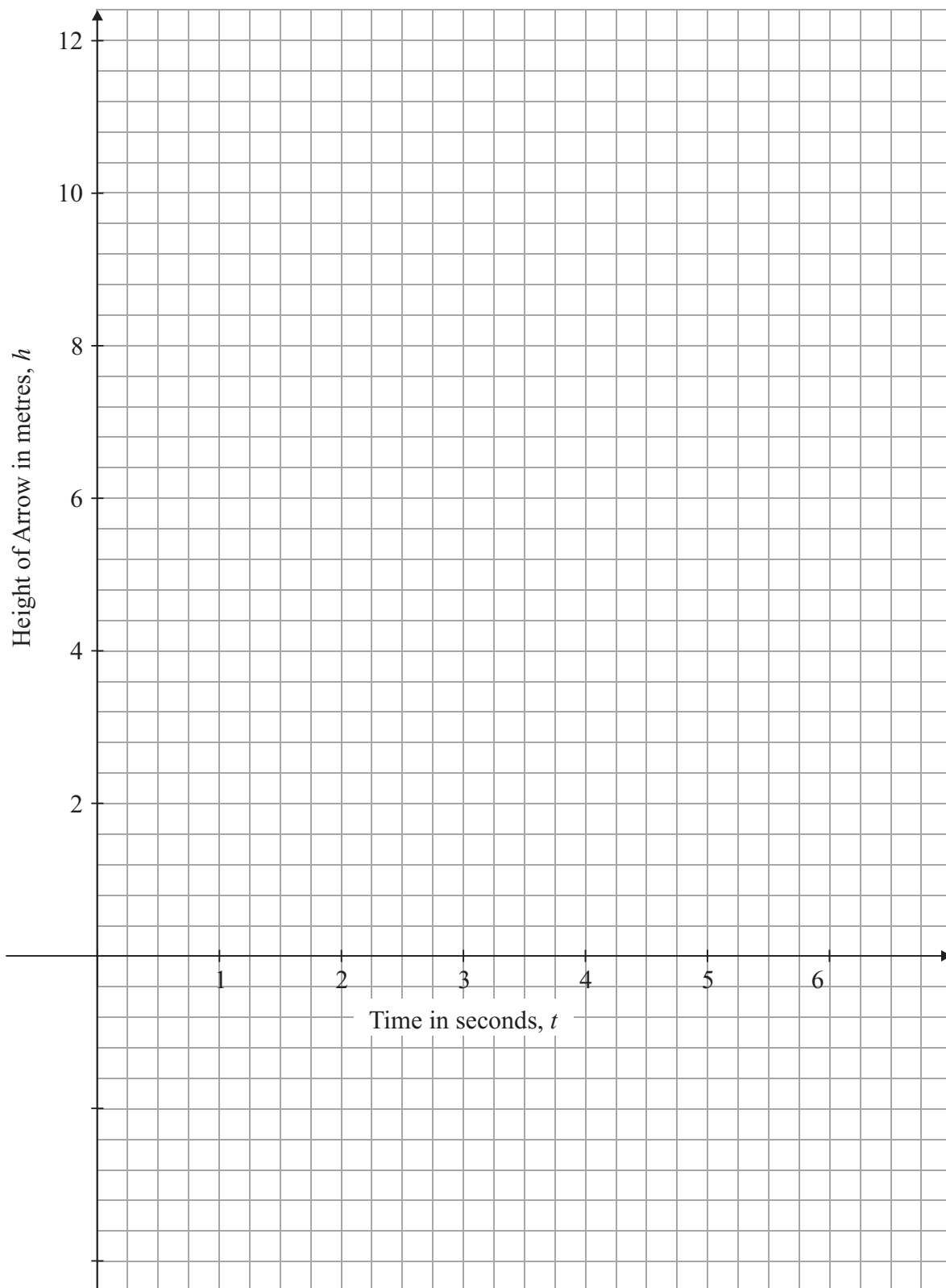
Question 12**(Suggested maximum time: 15 minutes)**

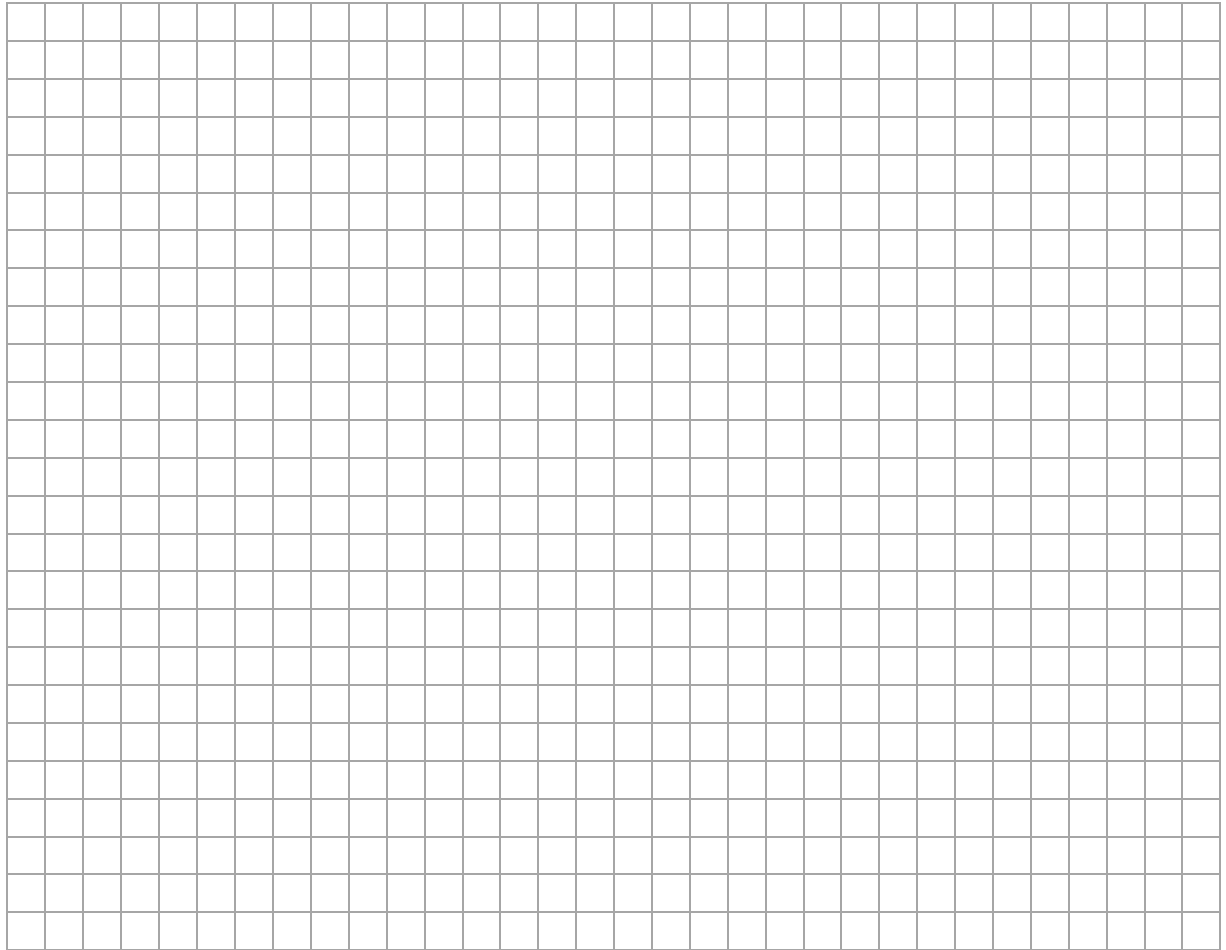
An archer, standing on level ground, shoots an arrow into the air. The height, in metres, of the arrow above ground level, t seconds after the arrow is released, is given by $h : t \mapsto 2 + 6t - t^2$.



- (a) On the grid below, draw the graph of $y = h(t)$ in the domain $0 \leq t \leq 6$, where $t \in \mathbb{R}$.

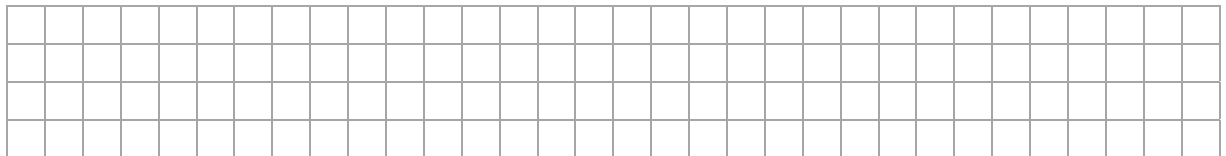
There is room for working out on the next page.



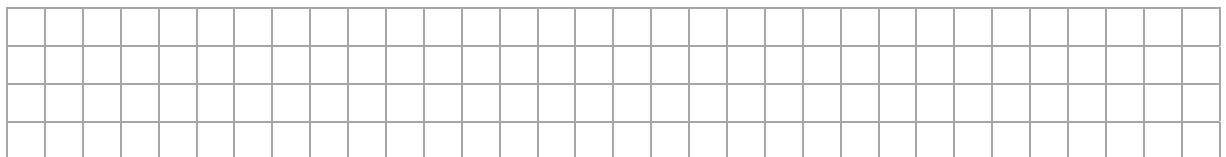


For parts **(b)**, **(c)** and **(d)**, you **must** show your working out on the diagram on the previous page.

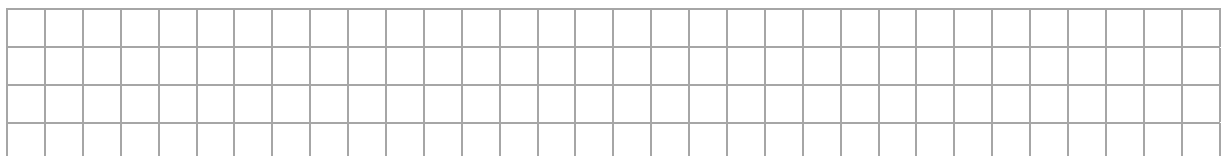
- (b)** Use your graph to estimate the height at which the speed of the arrow is zero after it is shot.



- (c)** Use your graph to estimate the length of time that the arrow is more than 6 m above ground level.



- (d)** Imagine the archer shoots the same arrow standing on an elevated position 3 m above ground level. By extending your graph, estimate the time that it would take the arrow to hit the ground.

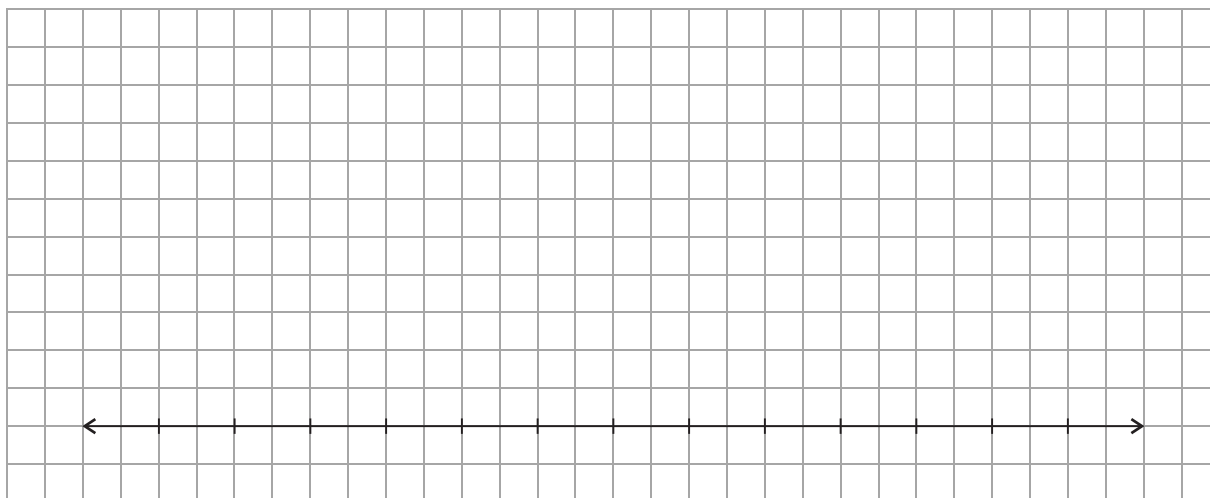


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Question 13**(Suggested maximum time: 10 minutes)**

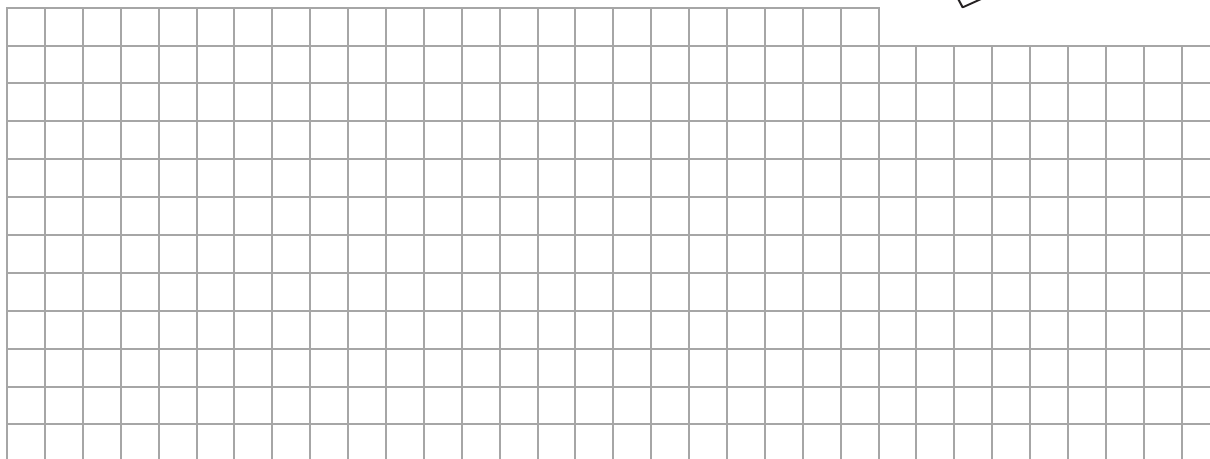
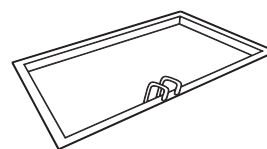
- (a) Solve the following inequality and graph your solution on the number line.

$$-6 < 4x + 2 \leq 14, x \in \mathbb{R}.$$



- (b) The area of a swimming pool is given by $2x^3 + 6x^2 - 25x - 25$.

- (i) The width of the swimming pool is $x + 5$.
Find the length of the pool, in terms of x .



- (ii) Given that the length of the swimming pool is 25 m, find the possible widths of the pool.
Which answer do you think is most appropriate? Give a reason for your answer.

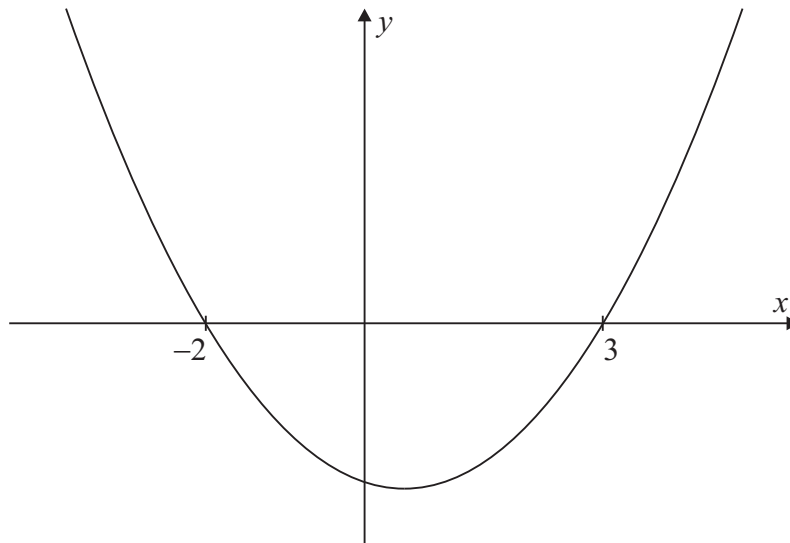


Question 14

(Suggested maximum time: 10 minutes)

The diagram below shows part of the graph of the function

$$f: x \mapsto x^2 + bx + c, \text{ where } x \in \mathbb{R} \text{ and } b, c \in \mathbb{Z}.$$



The graph of f intersects the x -axis at the points where $x = -2$ and $x = 3$.

- (a) Find the value of b and the value of c .

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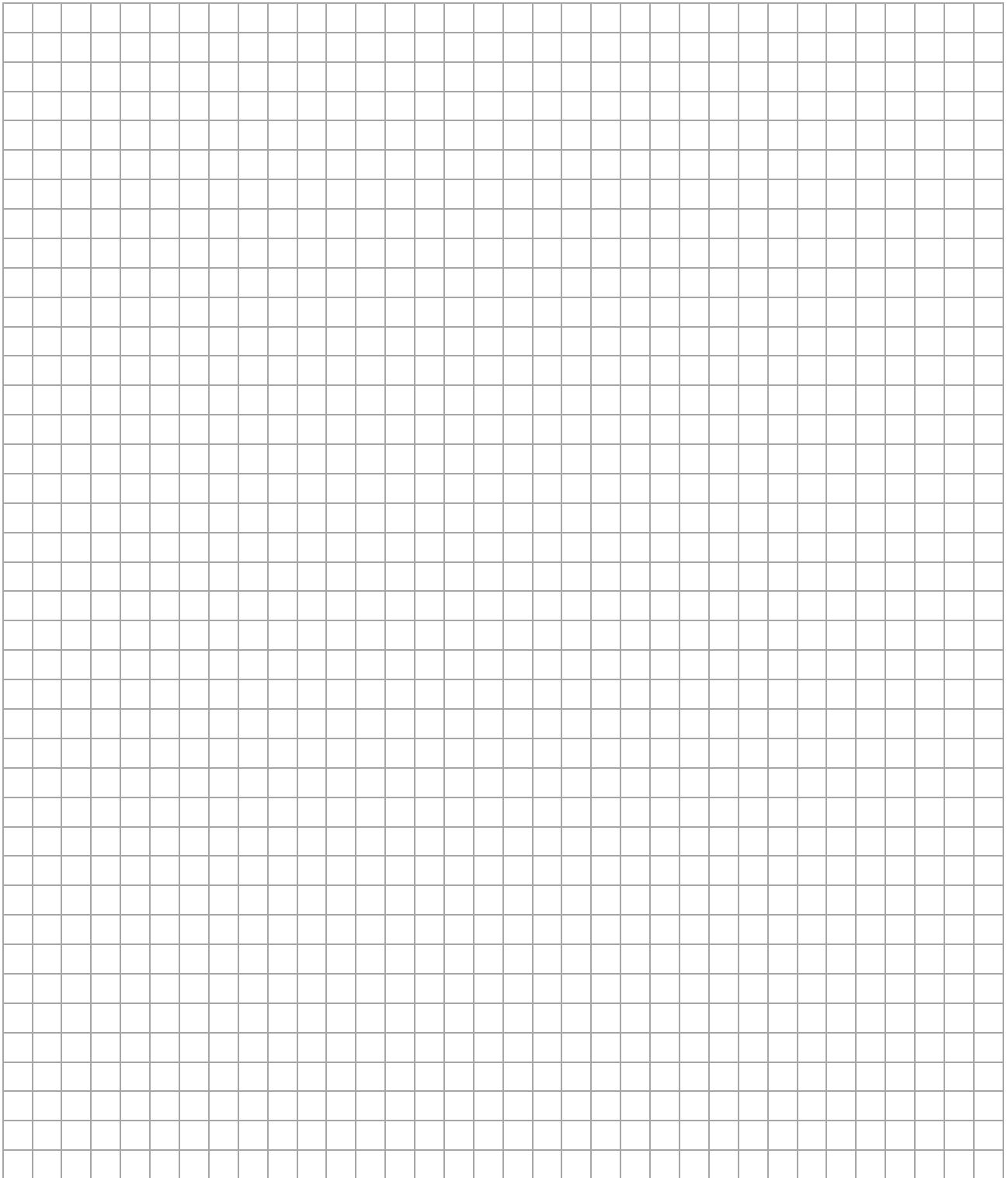
- (b)** Write down the co-ordinates of the point where the graph intersects the y -axis.

[illegible]

- (c)** $(t, 4t)$ is a point on the graph, where $t \in \mathbb{Z}$. Find the two possible values of t .

[illegible]

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Pre-Junior Certificate, 2016 – Higher Level

Mathematics – Paper 1

Time: 2 hours, 30 minutes

